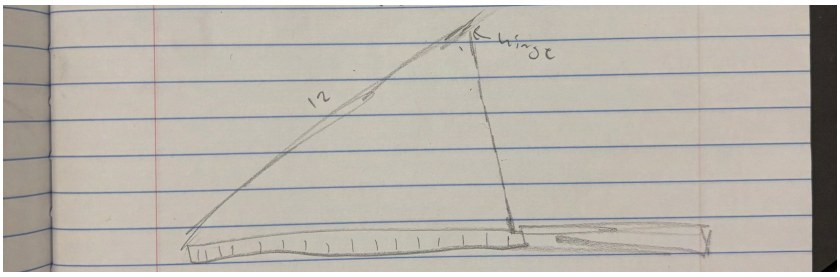
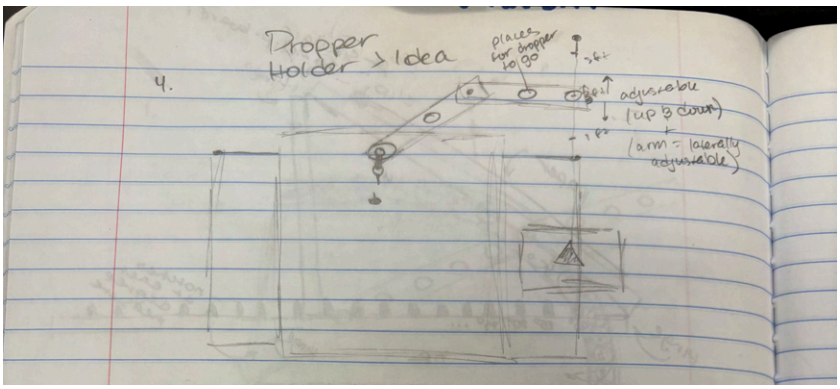
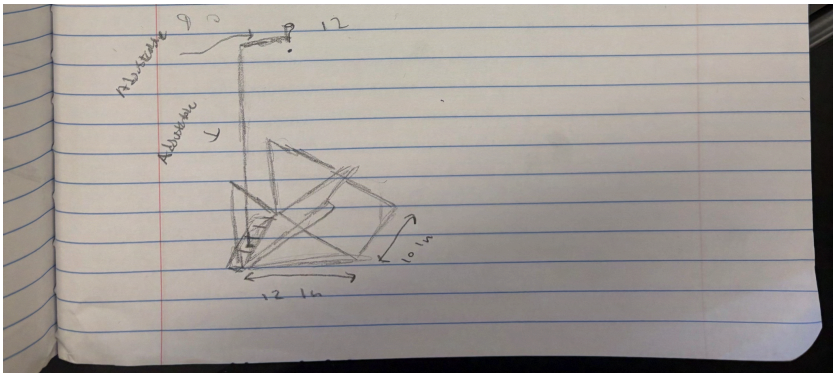
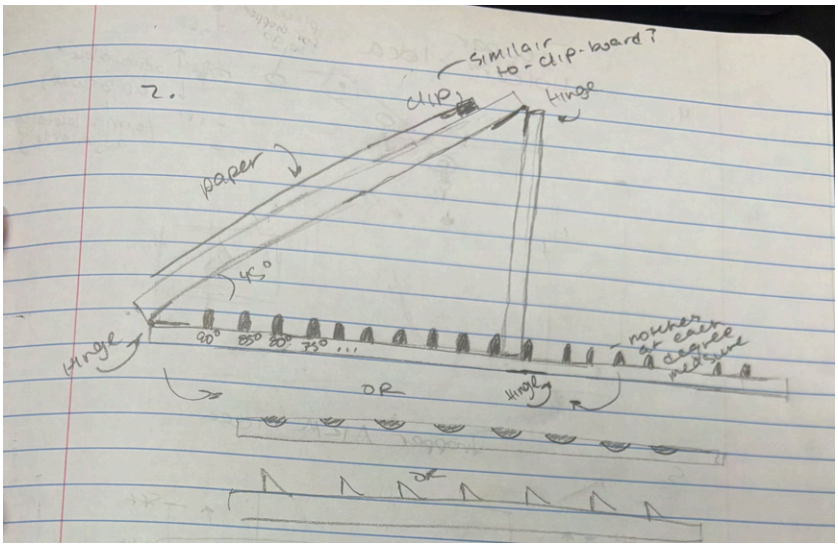
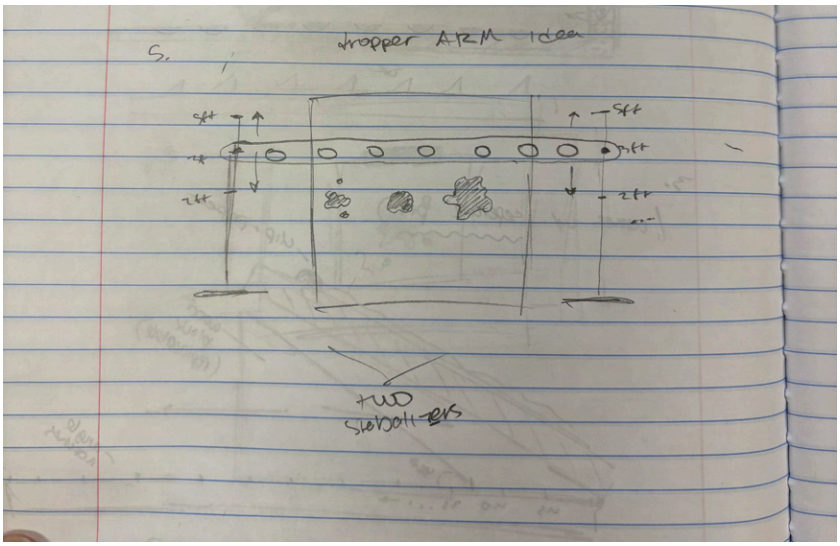
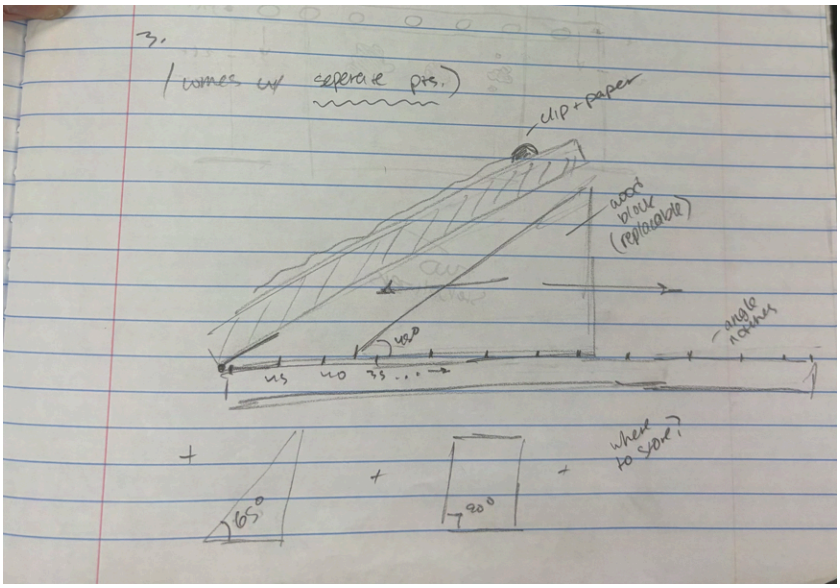
Element D	Information	Other Details	Date Added
D1	After much time was spent formulating ideas, some including “outlandish” and extremely interesting plausible solutions, we ended up with the following: 1.  2.  3.  4.		4/23/25


2.
similair to clip-board?
clip
hinge
paper
hinge
90° 85° 80° 75° ...
number at each degree measure
OR
hinge
OR

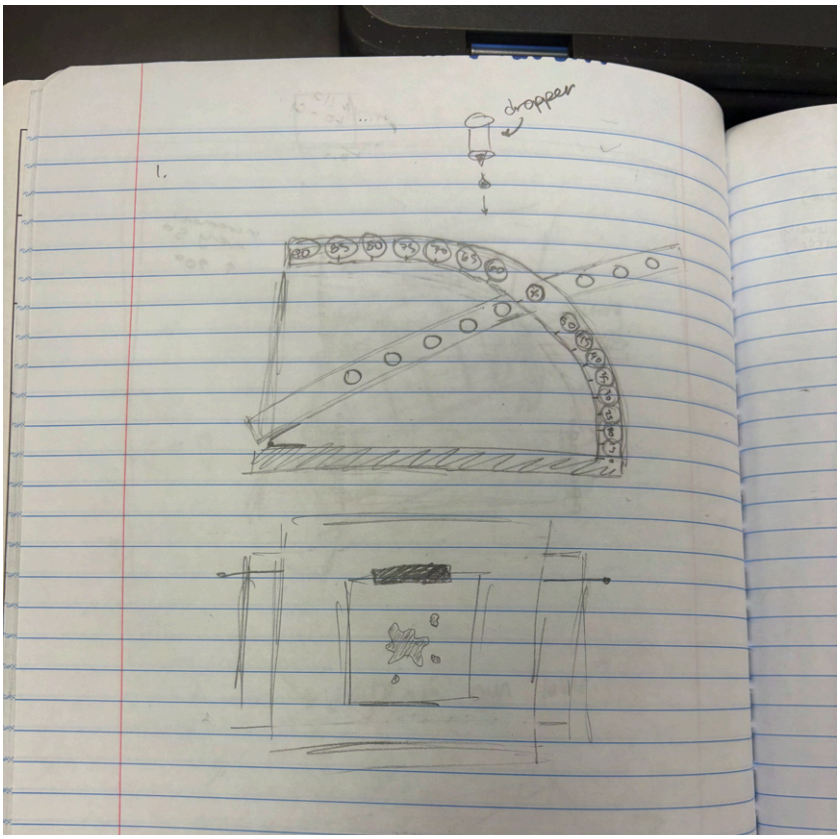
5.


5.
dropper ARM idea
5ft
5ft
2ft
2ft
2ft
two sidebarers

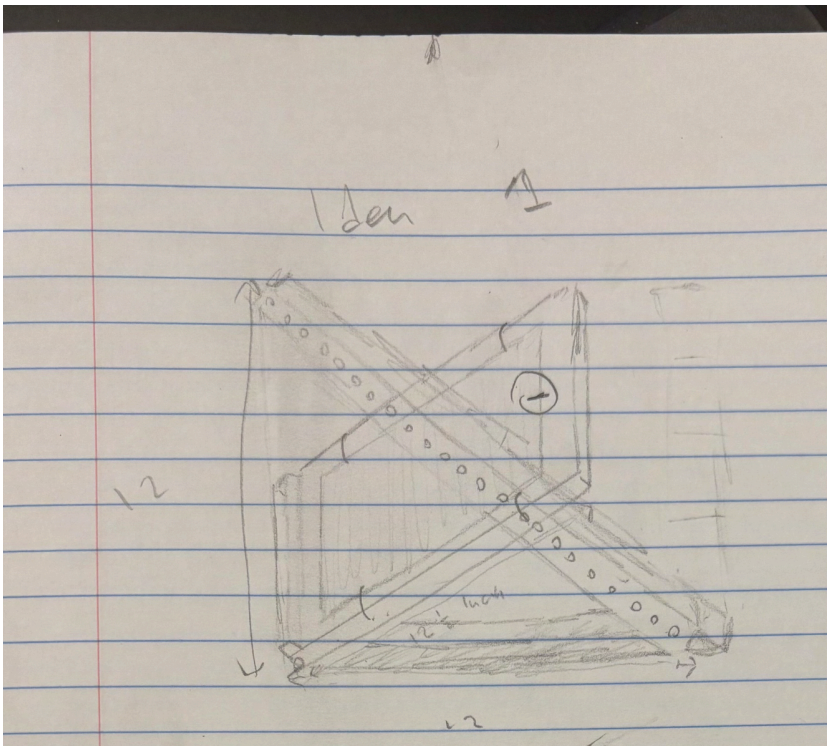
6.


3.
(comes w/ separate pts.)
clip+paper
wood block (replaceable)
angle markers
45 40 35 ...
65°
80°
where to store?

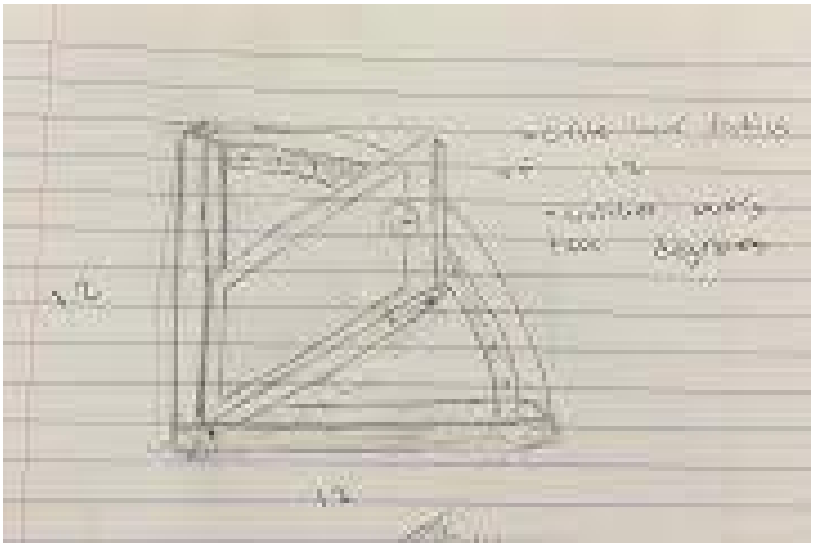
7.



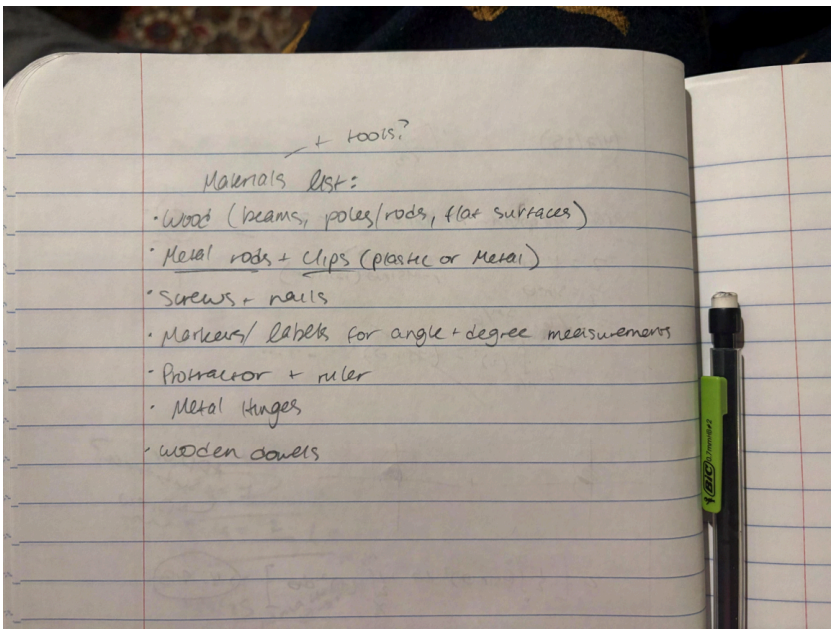
8.



9.

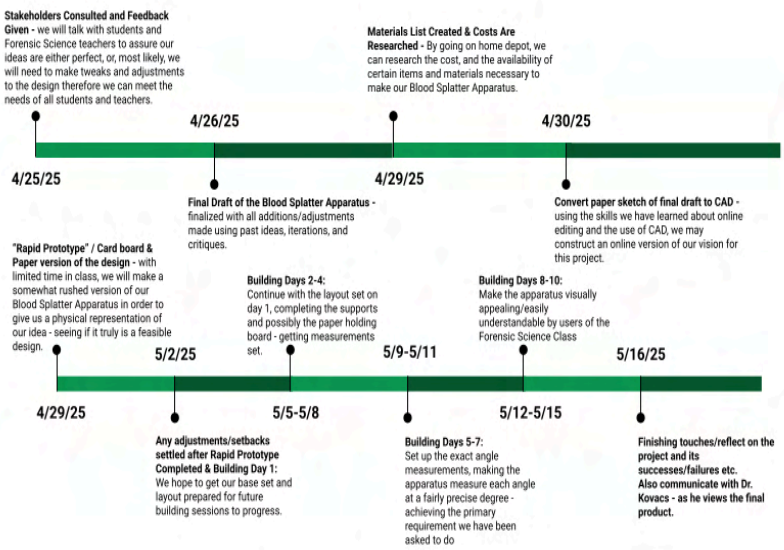


Idea 9
Writing says the following
-The circle has a radius of 12
Pins every 10 degrees
Works by inserting a pin through the board and quarter circle to set the board to the angle you want(in 10-degree increments)
The board flips up and down using a hinge.



Every idea is to be considered as possible, and iterative solutions to the problem of creating a functioning, reliable, and efficient system/apparatus that may allow students to accurately measure blood splatters during this “blood splatter analysis lab” in their Forensic Science Class.

D2	(*D2 Continued* may be seen at the bottom of this chart)		4/26/25

	We chose **idea 9** because it meets all of our criteria in the simplest and easiest manner. It serves its main purpose, to be an adjustable board that holds a standard piece of paper, and it also meets all of our other needs. Its simple design means its weight will be kept light, and therefore it will be moveable. The biggest pro to it is it is very easy to use and will prevent no challenges with the students using it. It will be durable as a result of its sturdy wood frame and its lake of an excess of movig parts. It will to meet the needs of our stakeholder DR Kovac.		
D3	 **(Action Plan Timeline)** How others will be able to reproduce results later as needed: By analyzing our detailed blueprint, copying the exact dimensions, and then building a replica to scale using the same materials list and all the same measurements. Following the Action Plan Timeline step by step will allow others will be able to consistently make multiples of the same device.		
D4	Students (some in the forensic science class, others signing up for the class in the future, and some just interested in the topic/problem we are facing): Certain questions about stakeholders/effects on a variety of people Liked the adjustable/slide mechanism Wondered if blood would be contained on the single piece of paper,		

given the height at which it is being dropped

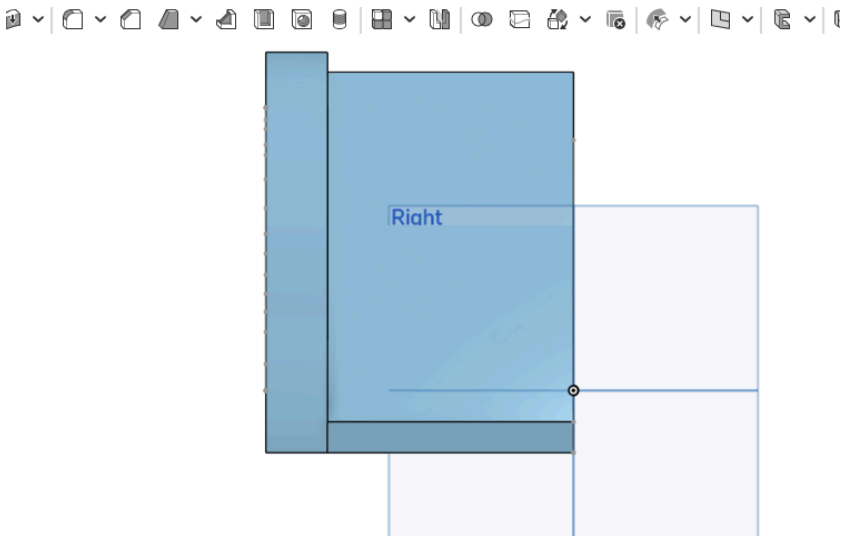
(A current forensic science student) Agreed that the lab, as it stands, is fairly limited, and enhancing the various mechanisms would likely expand the lab to include more detailed measurements/collecting of data.

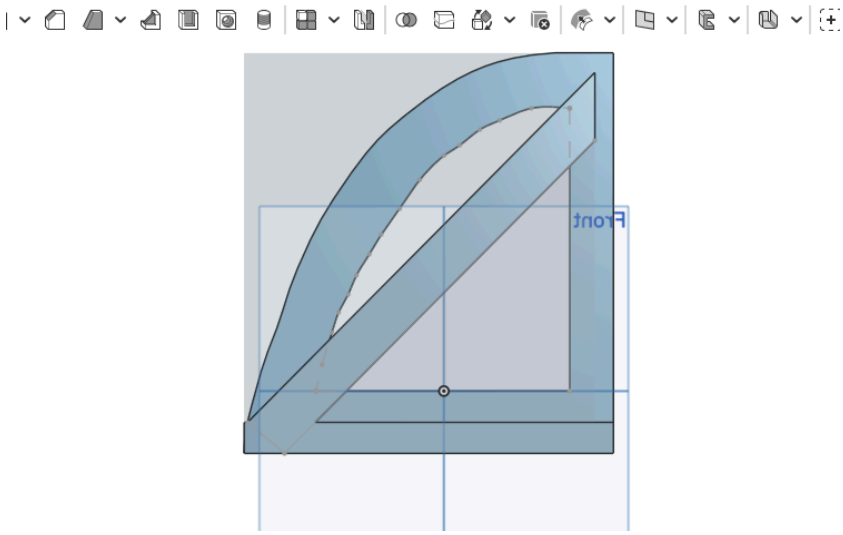
"Dr Kovac told us that our rapid prototype has "satisfied all my needs, and that dosent happenn to often." and "you have my aproval to move onto the next step."

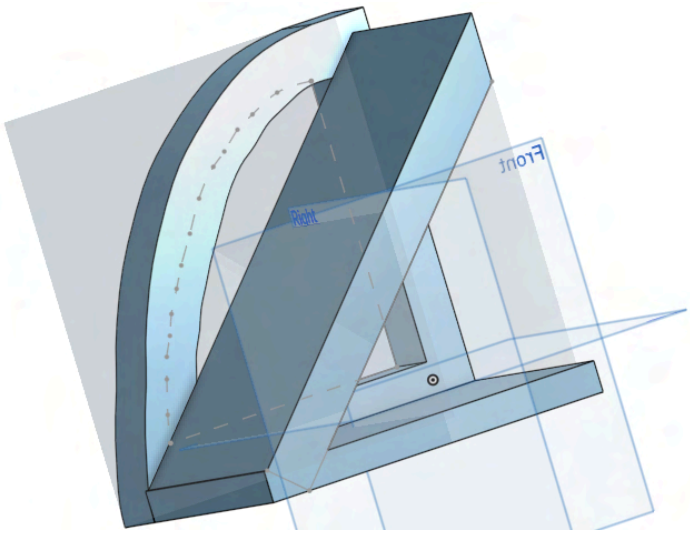
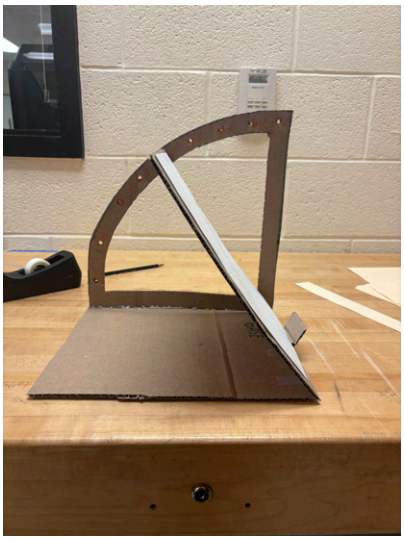
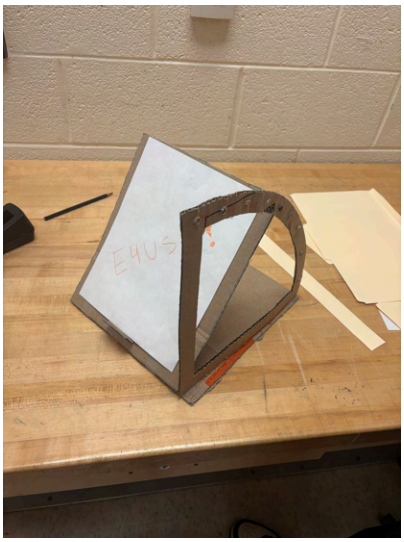
Item Name	Quantity	Thickness	Length	Width	link
Example: 2" x 4" x 96"	1	1 ½"	14"	3 ½"	LINK
1 in. x 2 ft. x 2 ft. Pine Edge Glued Panel Round Common Softwood Boards	1	1"	2'	2'	https://www.homedepot.com/p/1-in-x-2-ft-x-2-ft-Pine-Edge-Glued-Panel-Round-Common-Softwood-Boards-682527/202017012 $17.50
1 in. x 12 in. x 3 ft. Edge-Glued Pine Board	1	¾"	3'	12"	https://www.homedepot.com/p/Kimberly-Bay-1-in-x-12-in-x-3-ft-
Set of 4 Stainless Steel Flat Alligator Clips	1				https://www.homedepot.com/p/ExcelSteel-Set-of-4-Stainless-
					Steel-Flat-Alligator-Clips-635/319389575#overlay $5.60
Door Hinges for Interior Doors 3-Pack 3-1/2-inch x 3-1/2-inch, Design House 5/8-inch Radius Steel Door Hinge Door Hardware, Matte Black,	1	3-½ inch	3-½ inch	⅝ inch	LINK $7.35

3/16 in. x 3 ft. Zinc Plated Steel Solid Round Rod	1	3/16 inch	<1ft		LINK $5
					LINK
					LINK
					LINK
					LINK

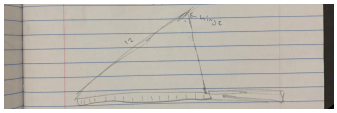
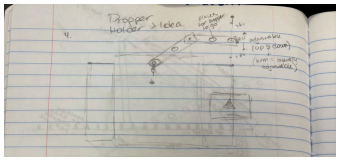
Total Price: $45.5

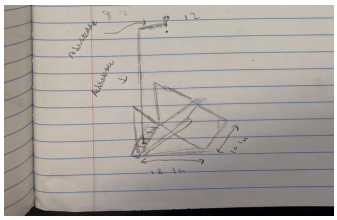
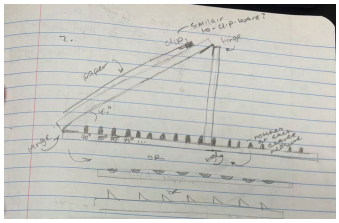


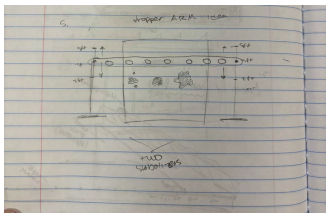
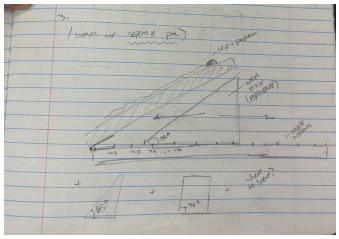
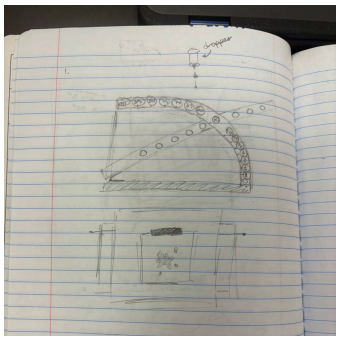


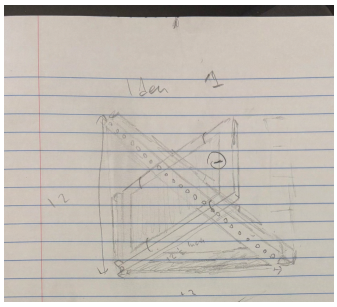
	  		

D2. (continued)

Idea/Concept for the project	Unique details about this idea	Build it? Yes/No	Date Added
Idea 1 	Idea 1 contains a reclining mechanism similar to those of beach chairs. This reclining mechanism is scaled down to fit our needs for specific angle and dimension requirements.	Yes, this is one of the more straightforward designs, as the mechanism is fairly manual and easy to use, therefore making it a feasible option.	4/25/25
Idea 2 	Idea 2 includes an additional aspect to the design which we had not yet thought of: a dropping mechanism that allows the student to 1. Adjust the angle of the paper and 2.	Somewhat possible, although the dropper-arm addition may disable our group from focusing on the main task at hand. This part may need to be a later decision if time permits,	4/25/25

	Maintain a perpendicular angle as the student changes the height at which the dropper is held. Having a mechanism to reduce possible error furthers the effectiveness of the lab in general. This specific design includes an adjustable “arm” with two hinges giving lateral advantage, while also being able to slide up and down-adjusting the height at which the blood is dropped as well.	otherwise, for the requirements Dr. Kovacs has outlined, it is unnecessary.	
Idea 3 	Idea 3 includes a “dual support system” on either side, as well as the dropper mechanism. The dropper in this case is a single arm, lending fewer advantages than other ideas, although it may be more structurally secure and less wobbly at certain heights.	Again, the dropper mechanism is not an absolutely necessary inclusion, although it would allow for more accurate measurements to be taken with a relatively simple addition to the design. Remember, however, that this entire apparatus should be able to be used by one or two students at a time; including more detailed functions may make the design too complex.	4/25/25
Idea 4 	Idea 4 displays a “foldable chair” esque design, with notches at every degree measure, with hinges attached to the two pieces of wood (the paper holder & supporting plank). There are possible additions/adjustments to the pegs- either making right triangles, or indentations into the wood to hold the supporting plank, are possible as well.	This design is fairly possible to build, as the support + peg system seems straightforward to construct within the given time constraint, and the variety of materials is quite small.	4/25/25
Idea 5	Idea 5 includes another “dropping mechanism” that displays an important aspect we may use for	In relation specifically to the blood dropper mechanism, as stated before, it is somewhat	4/25/25

	either the dropper or angle adjustment supports- having planks/supports on either side to stabilize the apparatus. With posts on either side of the board, there is less possibility of shaking and wobbling during use. This is an essential aspect due to our focus on accuracy while taking measurements. The dropper plate itself, which is held above the paper, has multiple holes where the dropper may be placed in order to adjust the droplet laterally across the paper, however an issue arises as this design limits the droppers ability to be dropped on anywhere other than the middle of the paper (cannot move up or down the paper)	unnecessary, although the idea includes key details and features which may be used for the paper holder and angle adjusting system as well, making these aspects possible to build in our given time constraints.	
Idea 6 	Idea 6 is somewhat unusual as it requires a sort of "block system" in which multiple pieces are brought as a "set" along with the single apparatus. It includes hinges connecting the paper-holding board and the surface board, as well as a fairly straightforward and easily understandable mechanism to adjust the angle.	Although the idea is fairly straightforward, the design itself includes far too many parts and can be seen as a blunt way to achieve the requirements with little thought into any other way of adjustment, therefore, although the construction of this idea is plausible, it is unlikely that we will carry through with this idea by itself.	4/26/25
Idea 7 	Idea 7 displays a separate approach to angle adjustment than has been seen before. This design includes a sort of curved support with angle markings along it, in which the paper-holding board may be connected using wooden dowels or rods, pushing them through both the angle measuring support and a hole in the	I believe that if we overlook the difficulty of creating curved wood supports, especially to the accuracy we would like this to be, the idea sounds great. The issue arises when creating an apparatus, with the skills we as student engineers have, that has shapes and measurements such as these. If we can overcome this issue, there are many	4/26/25

	wooden board. This idea includes supports on either side (angle markings may or may not also be on both supports) and is fairly compact, although the curvature of the supports may make construction fairly difficult/iterative.	great aspects to this design that we think could be generally beneficial to the apparatus as a whole.	
Idea 8 	Idea 8 is similar to previous ideas, in that it includes a "hole and dowel" sort of securing mechanism. A few downfalls occur when we observe the "one-sided nature" of this design. We want to ensure that the apparatus stays secure and does not wobble throughout the taking of measurements by any student. Having the post with holes in which we insert dowels/rods to connect the two pieces (the board and the support/post) only on one side may lend the mechanism to tilt or wobble. Furthermore, an issue occurs when we attempt to store or use this design if we were to build it in person, because the post with holes would have to be nearly twice as long, making it difficult to condense. The overall idea of this, however, seems fairly straightforward, and as said before, can be included in further designs.	Given time constraints and material constraints, this idea does seem like a possible design to construct, although due to certain flaws and possible improvements, we believe that certain ideas may be drawn from this design; however, Idea 8 alone will likely not be constructed with no adjustments or additions.	4/26/25